CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Forth Semester of B. Tech. Examination (EC)

November 2012

EC209 Microprocessors and Microcontroller

Date: 05.11.2012, Monday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 (a) What is a Microprocessor? What is the difference between a Microprocessor and Microcontroller? [03]

(b) List the addressing modes with suitable examples. [04]

Q - 2 (a) What is a flag? Explain each flag with help of examples. [05]

(b) Differentiate memory mapped I/O and I/O mapped I/O. [04]

(c) Draw the timing diagram for instruction IN 84H. [05]

OR

Q - 2 (a) Specify the contents of the registers and the flag status as the following each instructions are executed. [05]

i. MVI A, #00H
ii. MVI B, #F8H
iii. ADD B
iv. MVI D, #FFH
v. ADDC D

(b) Classify instruction set of 8085 microprocessor. [05]

(c) What do you mean by hardware interrupts? How many hardware interrupt supported by 8085? [04]

Q - 3 (a) Compare JMP and CALL instructions. [04]

(b) Differentiate between maskable and non-maskable interrupts. Also list the priority of each interrupts. [05]

(c) Explain the working of RST instructions and their use. [05]

OR

Q - 3 (a) What is stack memory? Explain operation of PUSH and POP instructions. [05]

(b) If the 8085 adds 87H and 79H, specify the contents of the accumulator and the status [03] of the S, Z, and CY flags.

(c) Discuss the programming model of 8085 microprocessor with the help of suitable diagram. [06]

SECTION – II

Q - 4 (a) List the Special Function Registers (SFRs) available in 8051 Microcontroller. [03]

(b) Explain following pin's function: [04]

i. ALE

ii. EA

iii. PSEN

iv. RST

Q - 5 (a) Write a 8085 assembly program to generate delay of 0.5 second. [07]

(b) Explain the memory organization of 8051 Microcontroller. [07]

OR

Q - 5 (a) Write a 8085 assembly program to find the square of the number from 0 to 9 using a table of Square. [08]

(b) Mention the ways B, C, D, E, H and L registers can be used. [06]

Q - 6 (a) Explain major components of 8259 with the aid of suitable diagram. [07]

(b) Interface 4-KB of memory with 8085 microcontroller with address range A000H-AFFFH. [07]

OR

Q - 6 (a) With a neat diagram describe the architecture of the 8085 interrupt system. [06]

(b) What is 8255? Discuss its various operating modes? [08]

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